

Class 12: Population Scale Analysis

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One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

How many samples do we have?

```
expr <- read.table("https://bioboot.github.io/bimm143_S26/class-material/rs8067378_ENSG00000172057.6.tx")
head(expr)
```

```
##      sample geno      exp
## 1 HG00367  A/G 28.96038
## 2 NA20768  A/G 20.24449
## 3 HG00361  A/A 31.32628
## 4 HG00135  A/A 34.11169
## 5 NA18870  G/G 18.25141
## 6 NA11993  A/A 32.89721
```

```
nrow(expr)
```

```
## [1] 462
```

The sample size for each genotype is shown below.

```
table(expr$geno)
```

```
##
## A/A A/G G/G
## 108 233 121
```

```
summary(expr)
```

```
##      sample      geno      exp
## Length:462    Length:462    Min.   : 6.675
## Class :character Class :character 1st Qu.:20.004
## Mode  :character Mode  :character Median  :25.116
##                                     Mean   :25.640
##                                     3rd Qu.:30.779
##                                     Max.   :51.518
```

The median expression levels for each genotypes are shown below.

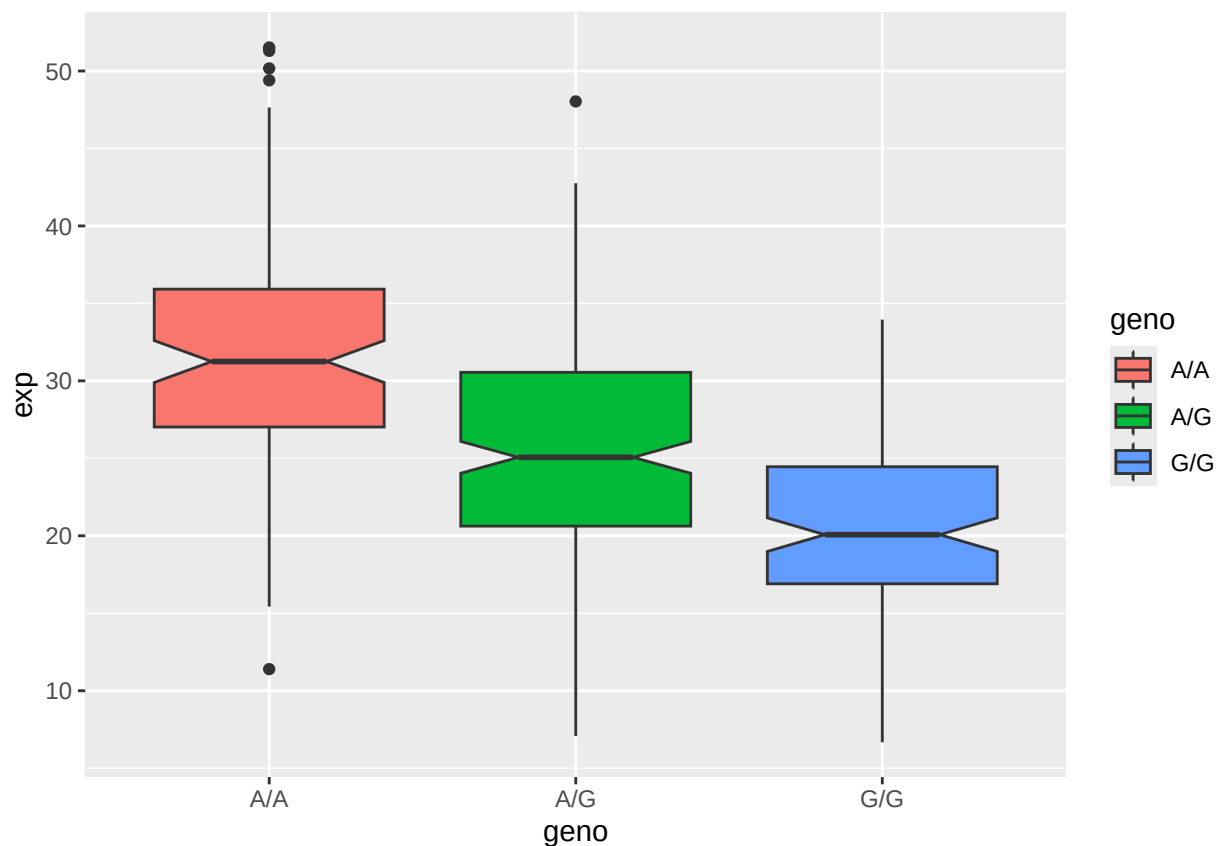
```
tapply(expr$exp, expr$geno, median)
```

```
##      A/A      A/G      G/G
## 31.24847 25.06486 20.07363
```

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

```
ggplot(expr) + aes(geno, exp, fill=geno) +  
  geom_boxplot(notch=TRUE)
```



The boxplot shows that the expression levels of ORMDL3 in G/G individuals are lower than that of A/A individuals. Thus, the SNP affects the expression of ORMDL3, more specifically by decreasing it.